

DEVANSH KALWANI

+91 8989734317 | devanshkalwani2010@gmail.com | github.com/devanshkalwani | linkedin.com/in/devanshkalwani

EDUCATION

Vellore Institute of Technology, Chennai B.Tech in Computer Science & Engineering (AI & Robotics) CGPA: 8.71	2028
Kautilya Senior Secondary School, Kota (CBSE) Class XII PCM 82.2%	2024
Macro Vision Academy, Burhanpur(CBSE) Class X 90.4%	2022

WORK EXPERIENCE

Python and FastAPI Development Intern, AltissAdvance Tech Pvt. Ltd. (Axentra OS) • Designed and developed RESTful APIs using FastAPI and Python for Axentra OS, collaborating with senior developers on new features, database integration (PostgreSQL/MySQL), and API documentation.	May 2026 – Aug 2026
---	---------------------

TECHNICAL SKILLS

Languages: Python, Java, C, C++

Web Development: HTML5, CSS3, JavaScript (ES6+), React, Django REST Framework, Node.js, Express.js

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs), REST APIs, WebSockets

Tools & Platforms: GitHub, Git, Linux, SQL Plus, SQLite3, Arduino, ESP32, ESP-NOW, Tailwind CSS

PROJECTS

MedConnect - Emergency Healthcare System | Python, Django 5.1, React, SQLite3, WebSockets, REST API

- Built a full-stack emergency clinical dispatch platform using Django 5.1, React, and WebSockets targeting 40,000+ underserved citizens in Tier-2/3 cities, with a novel Radially Expanding Dispatch (RED) algorithm that cuts emergency response latency by ~40% vs. static broadcast systems.
- Invented a HIPAA/ABDM Zero-Knowledge Dispatch protocol masking patient PII during hospital search and revealing EHR (blood group, allergies, chronic conditions) only on formal acceptance, creating a legally auditable dispatch trail.
- Delivered a dual-portal system - Patient Hub (one-tap SOS, digital EHR) and Clinical Command Center (live SOS queue, ICU bed tracking) - enforcing a 3-state lifecycle (SEARCHING -> ASSIGNED -> RESOLVED) with atomic-transaction concurrency control.

MineGuard Pro - Smart Safety Helmet System | ESP32, Arduino, C++, ESP-NOW, WebSockets, HTML/CSS/JS

- Engineered a real-time underground worker safety system using dual ESP32s over ESP-NOW (router-free mesh relay), achieving 1-second sensor telemetry and extended coverage across large environments with zero added infrastructure.
- Designed a priority-ranked C++ alert state machine (SOS > Fall > Gas > Temp > Sound > Heart) processing multi-sensor data in real time, ensuring zero missed critical alerts with per-alert buzzer patterns and physical ACK rescue button.
- Built a self-hosted HTML/CSS/JS supervisor dashboard served from ESP32 PROGMEM flash, featuring live WebSocket data push, real-time Chart.js trend graphs across 5 channels, and a 20-event incident history log - fully accessible offline via device Wi-Fi hotspot.

Smart OPD Queue Manager | React, Node.js, Express.js, REST API, Vite, Tailwind CSS

- Designed a full-stack OPD queue management system with a 3-tier priority engine (Emergency -> Priority -> Routine) and an aging promotion algorithm that auto-escalates routine patients to priority after 20 minutes of waiting, preventing queue starvation.
- Built a real-time Doctor Console with dispatch, pause/resume, and consultation-completion lifecycle controls backed by a stateful Express.js REST API, enabling zero-conflict single-consultation enforcement and millisecond-accurate wait-time tracking.
- Delivered an Operations Analytics dashboard computing live KPIs - average wait time, turnaround time, doctor utilization, and throughput - alongside a simulation engine for load-testing OPD flows with configurable dummy patient data.

CERTIFICATIONS

AI Foundations Associate - Oracle Gained foundational knowledge of cloud infrastructure and AI concepts for building scalable, intelligent solutions.	Oct 2025
Machine Learning with AI - Internshala Acquired hands-on exposure to ML model training, data preprocessing, and supervised learning for real-world prediction tasks.	Aug 2025

PROBLEM SOLVING & CODING

- 100+ DSA problems solved on LeetCode
- HackerRank Gold Badge - Python | HackerRank Silver Badge - C

LEADERSHIP & ACTIVITIES

- **Team Spirit** - Coordinated logistics for TechnoVIT'25 (Event Operations & Coordination Team), supporting smooth execution of technical events attended by 1,000+ participants.
- **Leadership** - Senior member of the Organizing Committee for Vibrance'26 (Discipline & Guest Care Committee), one of VIT Chennai's flagship annual cultural fests.